

Saber Ayatollahi, CFA, MSc

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SUMMARY

Model validation, model risk, and governance professional with ~7 years of experience across institutional ALM analytics, IRRBB modelling, and formal model-governance frameworks. At Moody's Analytics I hold delegated sign-off authority on valuation, sensitivity, and ALM outputs for multi-asset institutional portfolios (\$5-25bn per engagement), independently reviewing curve construction, spread calibration, cross-asset interactions, and assumption defensibility prior to production release. Built an enterprise cash-flow projection engine and stochastic scenario generators, each validated throughout the development lifecycle. CFA charter; dual MSc in Financial Modelling and Chemical Engineering.

CORE SKILLS

Model Validation & Independent Review • Model Risk & Governance (SR 11-7 / OSFI E-23 aware) • ALM & IRRBB Analytics • Assumption Validation & Escalation • Stress Testing & Scenario Analysis • Stochastic Modelling & Monte Carlo Simulation • Derivatives Sensitivity Validation • Python (pandas · NumPy · SciPy) • Cash Flow Projection Modelling

EXPERIENCE

Moody's Analytics – Asset Management & Buy-Side Solutions

Asst. Director – Modelling Services (Investment & ALM Risk Lead) | Toronto, May 2023 – Present

Model Governance & Independent Review

- Hold delegated sign-off authority on valuation, sensitivity, and ALM outputs for multi-asset institutional portfolios (\$5-25bn per engagement; ~\$50bn cumulative) – independently reviewing curve construction, spread calibration, and cross-asset interactions prior to production release.
- Operate within a formal model-governance framework: assess model risk by validating assumption defensibility, escalating outputs lacking mathematical or economic support, and maintaining documentation standards aligned with SR 11-7 and OSFI E-23 expectations.
- Serve as escalation point for client-critical analytical issues, interfacing directly with Heads of Risk and investment stakeholders on remediation plans.

Model Development & Validation

- Led design and implementation of an enterprise multi-asset cash flow projection engine supporting base, stress, and reverse-stress scenarios – embedded behavioral assumptions, prepayment logic, and macro stress overlays; validated outputs throughout the build.
- Validated derivatives pricing outputs across rates, FX, and inflation – confirmed sensitivity consistency and scenario impact integrity at portfolio-level ALM aggregates.
- Oversaw interest-rate risk and duration analysis under parallel and non-parallel rate shocks, aligned with IRRBB standards analogous to OSFI B-12 and Basel Committee frameworks.
- Migrated manual spreadsheet valuation workflows into auditable Python analytics pipelines; ran both environments in parallel for two production cycles to confirm output reconciliation before cutover – governance audit closed satisfactorily.

Tooling & Efficiency

- Built agentic AI development workflows (Claude Code, Cursor IDE) automating first-pass code review, validation scaffolding, and anomaly detection – reducing development cycle time by an estimated 30-40% on comparable modules while preserving human sign-off on governance-critical steps.

Client Service Specialist – Investment Analytics | Toronto, May 2022 – May 2023

- Led onboarding and platform implementation for U.S. and Canadian pension funds, asset managers, and consulting firms migrating to the PFaroe analytics platform – validated model outputs post-deployment and translated client investment-team requirements into structured Product Owner requests.
- Successfully migrated all assigned client accounts from a legacy Calypso environment to the PFaroe PM platform.

Ernst & Young (EY)

Senior Consultant – IFRS & Financial Risk Transformation | Toronto, Sep 2021 – Apr 2022

- Delivered IFRS 17 and IFRS 9 technology-transformation projects for Canadian and international insurance clients – contributed to governance documentation, regulatory-readiness processes, and cross-functional alignment across actuarial, finance, IT, and risk teams.

Ortec Finance Canada

Consultant – Asset-Liability Management & Risk | Toronto, Feb 2019 – Sep 2021

- Built and interpreted stochastic economic scenario generators assessing interest-rate risk and funding volatility under base-case and stressed environments for Canadian and international pension-fund clients.
- Performed asset-only and asset-liability (surplus) portfolio optimization using VaR and CVaR via Ortec's GLASS platform – with risk decomposition, contribution-to-risk attribution, and near-optimal frontier analysis.
- Delivered ALM studies analyzing funding ratios, duration mismatch, and long-term balance-sheet risk; conducted cash flow, liquidity, and hedging analysis across interest-rate, inflation, and currency exposures.
- Led the ALM study for a three-plan university pension merger (new jointly-sponsored plan) – on a team of three developing and validating the model capturing all Funding Policy dynamics, evaluating duration, currency, inflation, and leverage overlay strategies.
- Presented ALM findings and strategic asset allocation recommendations to investment committees at Canadian pension-fund clients.

EDUCATION & CREDENTIALS

- Chartered Financial Analyst (CFA) – CFA Institute, 2024
- MSc, Financial Modelling – Western University, 2018
- MSc, Chemical Engineering (Full Scholarship) – Western University, 2016